Sleep continual learning: retention, consolidation and matched baselines

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Strengths and best use

Sleep addresses a concrete continual-learning problem by separating fast memory from slower consolidation. The researchers describe knowledge seeding and a synthetic-practice phase, then evaluate retention and new-knowledge incorporation. This offers a research design for testing how learning changes over time. Paper (https://arxiv.org/html/2606.03979v2) - Google Research overview (https://research.google/blog/the-sleep-framework-a-new-way-for-llms-to-retain-knowledge-in-continual-learning/)

Independent checks

The paper's factual-incorporation table reports 31.9% for the base model and 48.9% for Sleep on the single-passage SQuAD setting. Removing dreaming gives 35.7% in that comparison. Those are separate experimental conditions that should retain their labels. The blog describes an offline process within human-defined training rules. HOPE is related architecture research, while Sleep specifies a consolidation and rehearsal lifecycle. HOPE and Nested Learning (https://arxiv.org/abs/2512.24695)

Plan a small reproduction

1. Pin the paper version, model revision, data sequence, memory capacity, and training budget.
2. Evaluate old knowledge and new knowledge after every stage, including a held-out transfer slice.
3. Compare sequential training, replay, consolidation, and a consolidation-without-dreaming ablation under matched resources.
4. Measure forgetting, accuracy, compute, storage growth, and variance across seeds.
5. Keep synthetic curriculum generation separate from the held-out evaluation set. A budgeted pilot can test the method before attempting the full architecture.

Reusable prompt

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Turn this continual-learning result into a reproduction plan. Preserve dataset, model, and evaluation labels. Specify matched-budget baselines, retention after every stage, a dreaming ablation, seed variation, and compute and storage accounting. Mark implementation details that the available source does not resolve.
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Project idea

Build a continual-learning evaluation notebook with a retention matrix, baseline comparison, and resource ledger. Start with a small synthetic task sequence and fixed held-out questions.

Original sources

- https://www.instagram.com/reel/DfZuDIUOGJZ/